2. $g=7$ $p=23$ $x=3$ $y=5$

Shared symmetric key using the Diffie Hellman protocol is obtained by exchanging the values and taking modulus till a same value is reached. That is the shared key.

$7 \bmod^5 =$ $23 \bmod^3 =$

3. Registration Authority is where the nodes share the key and gets verified or they get the key of the other node.
It relates to Certificate Authority (CA) in the sense that RA is needed to issue a digital certificate verify the ~~config~~ confidentiality.

4. Alice's requirements is that the full message transfer ~~presses~~ process should satisfy integrity. There should be no tampering of the message. It can be acheived by encrypting the message and adding digital certificates to record if any tampering is done.

6. The 4 phases of the virus are:

- Stand by mode – is in passive mode until the user opens it.
- Active mode – when the user opens the file, it becomes active and starts running the code.
- Execution phase
- Passive phase.